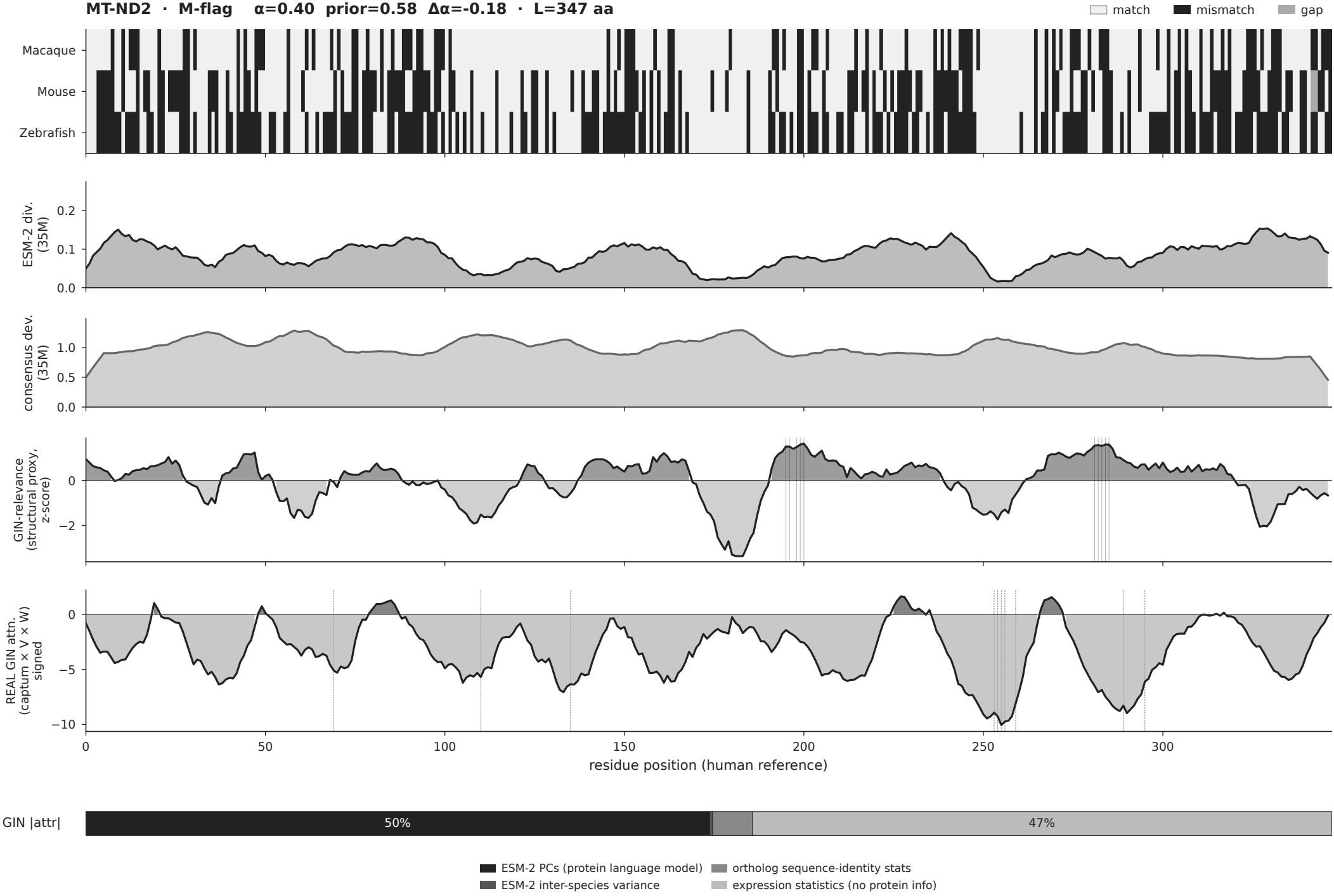
MT-ND2 · M-flag α=0.40 prior=0.58 Δα=-0.18 · L=347 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.0
0.1
0.2
consensus dev. (35M)
0.0
0.5
1.0
GIN-relevance (structural proxy, z-score)
0
−2
REAL GIN attn. (captum × V × W) signed
0
−5
−10
0
50
100
150
200
250
300
residue position (human reference)
GIN |attr|
50%
47%
ESM-2 PCs (protein language model)
ESM-2 inter-species variance
ortholog sequence-identity stats
expression statistics (no protein info)